International Journal of Heat and Fluid Flow 119 (2026) 110251

Contents lists available at ScienceDirect

International Journal of Heat and Fluid Flow

journal homepage: www.elsevier.com/locate/ijhff

The optimized design and heat transfer characteristics of helical groove tubes: A study based on parameter estimation, continuous adjoint optimization, and physics-informed neural networks

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ARTICLE INFO

Keywords:
Novel gas-phase rotary tube-shell heat exchanger
Heat exchange tube
Parameter estimation
Continuous adjoint optimization
Physics-informed neural network
Energy efficiency

ABSTRACT

To enhance the heat transfer performance of heat exchange tubes and address the research gap in the coupling of structural parameters, operating parameters, and energy efficiency indicators between Parameter Estimation (PE) and Continuous Adjoint Optimization (CADJ) in heat exchange tubes, this study focuses on the helical groove tube in a novel gas-phase rotary shell-and-tube heat exchanger. The study sequentially applies PE and CADJ methods for optimization design, resulting in a new type of highly efficient heat exchange tube. Based on the thermal performance-to-pressure drop loss ratio, Computational Fluid Dynamics software is used to quantitatively assess the heat exchange tube's energy efficiency and systematically analyze its heat transfer characteristics. Additionally, a physics-informed neural network (PINN) is employed to solve the Navier-Stokes equations and reconstruct the two-dimensional temperature field, thereby cross-validating the CFD results in the absence of experimental validation and enhancing the robustness of the optimized design conclusions. The results indicate a strong correlation between various parameters, including tube inner diameter, groove depth, pitch, air velocity, inlet fluid temperature, and tube wall temperature, and the heat transfer characteristics, with correlation coefficients of 0.9799, 0.9957, 0.9897, and 0.9989, respectively. It was found that enhancing the Nusselt number comes at the cost of increased pressure drop. Compared to the helical groove tube, the novel heat exchanger tube exhibits superior performance in both heat transfer efficiency and energy efficiency, with improvements in the Nusselt number, pressure drop, and the ratio of thermal efficiency to pressure drop by 22.03 %, 22.88 %, and 58.51 %, respectively. The superior performance of the new heat exchanger tube is attributed to the continuous optimization of the helical groove tube's inner wall morphology using the CADJ method, which maintains the basic structure of the internal helical ribs. This optimization strengthens the vortex flow structure while effectively preserving the smooth flow path of the fluid. Additionally, in the absence of experimental validation, the discrepancy between the PINN-reconstructed temperature field and the CFD solution remains on the order of 10^{-1}, indicating that employing PINN for cross-validation and auxiliary assessment of CFD results provides a simple yet efficient alternative approach. This study presents a reasonable, novel, and practical optimization strategy to improve the heat transfer performance and energy efficiency of heat exchange tubes, offering significant practical application value.

1. Introduction

Heat exchangers are a crucial component in thermal energy conversion systems, and heat exchange tubes, as the key heat transfer elements within them, play a significant role (Kalendar et al., 2011; Ahmad et al., 2012; Wu et al., 2018). The gas-phase rotary shell-and-tube heat exchanger, as a novel heat exchange device, utilizes a core heat transfer element–the helical groove tube–that plays a vital role in energy conservation and consumption reduction during the grain drying process. The helical groove tube, invented and manufactured by the American company *Pabco* in 1956, features a unique groove structure on the tube

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https://doi.org/10.1016/j.ijheatfluidflow.2026.110251
Received 11 October 2025; Received in revised form 12 December 2025; Accepted 5 January 2026
Available online 9 January 2026
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Nomenclature

c_p	Specific heat capacity (at constant pressure), J•(kg•K)$^{-1}$
d_{in}	Inner diameter of groove tube, mm
d_{out}	Outer diameter of helical groove tube, mm
e	Groove depth, mm
f	Darcy resistance coefficient, Dimensionless number
L_{eff}	Helical groove tube length, mm
$\dot{m}_{inlet}$	Inlet mass flow rate, kg·s^{-1}
n	Number of groove
Nu	Average Nusselt number, Dimensionless number
p	Pitch, mm
P_{inlet}	Inlet pressure relative to atmospheric pressure, Pa
P_{outlet}	Outlet pressure relative to atmospheric pressure, Pa
Pr	Prandtl number, Dimensionless number
q	Heat flux density, W•m^{-2}
Re	Reynolds number, Dimensionless number
R	Thermal performance-pressure loss ratio, Pa^{-1}
T_{fluid}	Fluid temperature, K
T_{inlet}	Inlet fluid temperature, K
T_{wall}	Wall temperature, K
v	Air velocity, m•s^{-1}
w	Groove width, mm
y^+	Distance of the viscous sublayer near the wall, mm
ΔP	Pressure loss, Pa
Δy	Distance from the first mesh layer to the wall, mm

Greek symbols

β	Helix angle, °
λ	Thermal conductivity, W•(m•k)$^{-1}$
μ	Viscosity coefficient, N•s•m^{-2}
ρ	Fluid density, kg•m^{-3}
$\|\omega\|$	Vorticity magnitude, s^{-1}
τ_w	Wall shear stress, N

Superscript

n	Number

Subscripts

eff	Effective
in	Inner parameter
i,j,k	Coordinate directions
inlet	Inlet parameter
out	Outer parameter
outlet	Outlet parameter
p	Constant pressure condition
wall	Wall parameter
fluid	Fluid

wall, which induces strong rotation and turbulence in the fluid flow (Naphon and Wongwises, 2004; Pan et al., 2025; Jeon et al., 2025). This structure effectively enhances heat transfer performance and significantly improves heat exchange efficiency. Extensive research has been conducted, both domestically and internationally, on improving the heat transfer characteristics of these tubes through structural optimization methods. These studies primarily focus on two directions: gradient-free optimization and gradient-based optimization. The gradient-free optimization methods include parameter estimation, gradient-free optimization algorithms, and gradient-free machine learning techniques, which do not rely on gradient information. On the other hand, gradient-based optimization methods, such as sequential quadratic programming and the steepest descent method, use gradient information to determine the search direction, leading to a significantly faster convergence rate compared to gradient-free methods.

In recent years, gradient-free optimization methods based on Computational Fluid Dynamics (CFD) have been widely applied in the optimization design of heat exchangers, particularly demonstrating significant advantages in the optimization of different heat exchanger tube structures. For example, the research team led by L. Wan and colleagues (Wan et al., 2024; Wan, 2016; Wan, 2016) has focused on the geometric configuration optimization of key heat transfer elements in gas-phase rotating shell-and-tube heat exchangers. The team utilized both experimental performance tests and numerical simulation methods to systematically analyze the influence mechanisms of the geometric structures in both the shell-side and tube-side on the heat transfer characteristics (Diao et al., 2019; Zhang et al., 2020; Tian et al., 2024). During their research, they employed parameter estimation methods such as the Box-Behnken design and Response Surface Methodology (RSM) to construct a multi-dimensional design space, revealing the coupling relationships between structural parameters and their synergistic effects on heat transfer performance. Additionally, the team incorporated various mathematical fitting models to enhance function approximation, streamline the design process, and improve computational efficiency (Zheng et al., 2025). P. Ranut and other researchers (Ranut et al., 2014) applied Multi-Objective Genetic Algorithms (MOGA) to optimize the shape of tube heat exchangers, combining CFD-based surrogate modeling to significantly reduce the overall computational cost during the optimization process. In the same study, D. Bacellar and colleagues (Bacellar et al., 2017) introduced the Kriging model as a surrogate model, suitable for scenarios with fewer structural parameters, thus reducing optimization time. X. Gu and others (Gu et al., 2018) employed multi-parameter optimization methods to systematically analyze the relationship between structural parameters, such as the inclination angle, width, spacing, and number of baffles, and the heat transfer characteristics of twisted flow heat exchangers. Their results demonstrated that the baffle inclination angle and spacing have the most significant impact on heat transfer performance, followed by baffle width, while the number of baffles had the least effect. S. Fettaka and colleagues (Fettaka et al., 2013) utilized Genetic Algorithms (GA) to study the detailed design of shell-and-tube heat exchangers, proposing a new multi-objective optimization algorithm aimed at balancing heat transfer area and pump power consumption. S.K. Sarangi and his team (Sarangi et al., 2021) studied curved trapezoidal fins, selecting arc radius, angle, and fin height at the large end as optimization variables. They applied Latin Hypercube Sampling (LHS) and CFD simulations, using Backpropagation Neural Networks (BPNN) and GA for multi-objective optimization, proposing a design that balances thermal performance and energy efficiency. H.D. Li and others (Li et al., 2024) proposed an improved intelligent evolutionary algorithm, combining multi-objective functions and discrete variables, for the optimization of shell-and-tube heat exchanger design.

For optimization problems with the large number of design variables, gradient-based methods, particularly those using adjoint optimization to calculate sensitivities, are more efficient (Mathesen et al., 2021). The adjoint method efficiently calculates the gradient of the objective function with respect to all design variables by solving the adjoint equations related to the objective function, requiring only one computation that is roughly equivalent in cost to solving the flow field once. This method is especially suitable for optimization problems with multiple design variables (Jameson, 1988; Nadarajah and Jameson, 2001). Compared to gradient-free methods, the adjoint method generally requires fewer iterations, leading to faster convergence during the optimization process. R. Zhang and others (Zhang and Qian, 2020) conducted a non-parametric shape optimization of various fin structures, using surface grid nodes as design variables. The results indicated

that, under the same pressure drop as the reference design, the outlet temperature increased by 75 %. G.V. Stratan and colleagues (Stratan et al., 2024) experimentally confirmed the effectiveness of the adjoint method in the optimization of wind turbine blades. K. Wang and others (Wang et al., 2022) used adjoint optimization methods to perform three-dimensional non-parametric shape optimization of fins in a supercritical carbon dioxide Brayton cycle regenerator. M.C. Vidya and colleagues (Vidya et al., 2018) applied adjoint optimization to two-dimensional pin–fin arrays in a heat exchanger under steady laminar conditions, showing an 11 % reduction in pressure drop and an 11 % improvement in heat transfer performance. S. Sun and others (Sun et al., 2020) utilized continuous adjoint methods to optimize the shape of fins inside a three-dimensional radiator, where the optimized structure, consisting of two unconventional geometries, exhibited superior heat transfer characteristics. A. Ghasemi and colleagues (Ghasemi and Elham, 2021) coupled finite element models with adjoint solvers to optimize the three-dimensional fin shape, targeting the downstream air temperature of the radiator. The use of a high-aspect-ratio wing structure, with an outward curve from the base, not only effectively reduced flow resistance but also achieved an 18 % reduction in thermal resistance while maintaining the same pressure drop as the reference design.

In summary, existing optimization methods are primarily categorized into gradient-based and gradient-free approaches, with most relying on single optimization techniques. These methods remain relatively simple, and a comprehensive multi-strategy approach is yet to be established. Furthermore, research on the optimization design of helical groove tubes under pure gas-phase conditions is still limited. Therefore, this study adopts a combined strategy of "parameter estimation and continuous adjoint optimization" for the optimization design of helical groove tubes, aiming to develop a novel and efficient heat exchange tube to enhance its thermal performance. This approach also fills the gap in the application of multi-strategy combinations for coupling the structural, operating parameters, and energy efficiency indicators in heat exchange tubes. Additionally, given the geometric complexity of the optimized heat-exchange tube and the time-consuming nature of fabrication and experimental validation, a PINN is first employed to verify physical consistency and obtain high-resolution predictions of its flow and heat-transfer performance under prescribed operating conditions, thereby guiding subsequent specimen design and PIV experiments. Under this overall framework, the main methodological contributions of this work are as follows:

(1) An integrated optimization framework based on parameter estimation and continuous adjoint optimization (PE-CADJ) is developed for helical groove tube geometry and operating conditions, enabling the coordinated design of geometric configuration, structural parameters, operating parameters, and energy-efficiency metrics, while performing fine-scale inner-wall shape optimization under the constraint of preserving the helical rib, thereby achieving a more favorable trade-off between heat-transfer enhancement and pressure-drop penalty.

(2) A PINN is introduced to reconstruct the temperature field and to cross-validate CFD predictions under explicit physical constraints, providing a practical and efficient surrogate approach for assessing the reliability of numerical results when experimental measurements are not available.

2. Methodology description

2.1. Numerical simulation method

2.1.1. Physical model and fluid properties

This study focuses on the helical groove tube from the gas-phase rotary shell-and-tube heat exchanger developed by our research team. Due to the relatively large original dimensions of the helical groove tube, a tube with an effective length (L_{eff}) of 1000 mm was selected for numerical simulation. The experimental platform for the heat exchanger is shown in Fig. 1, and the specific structural dimensions of the helical

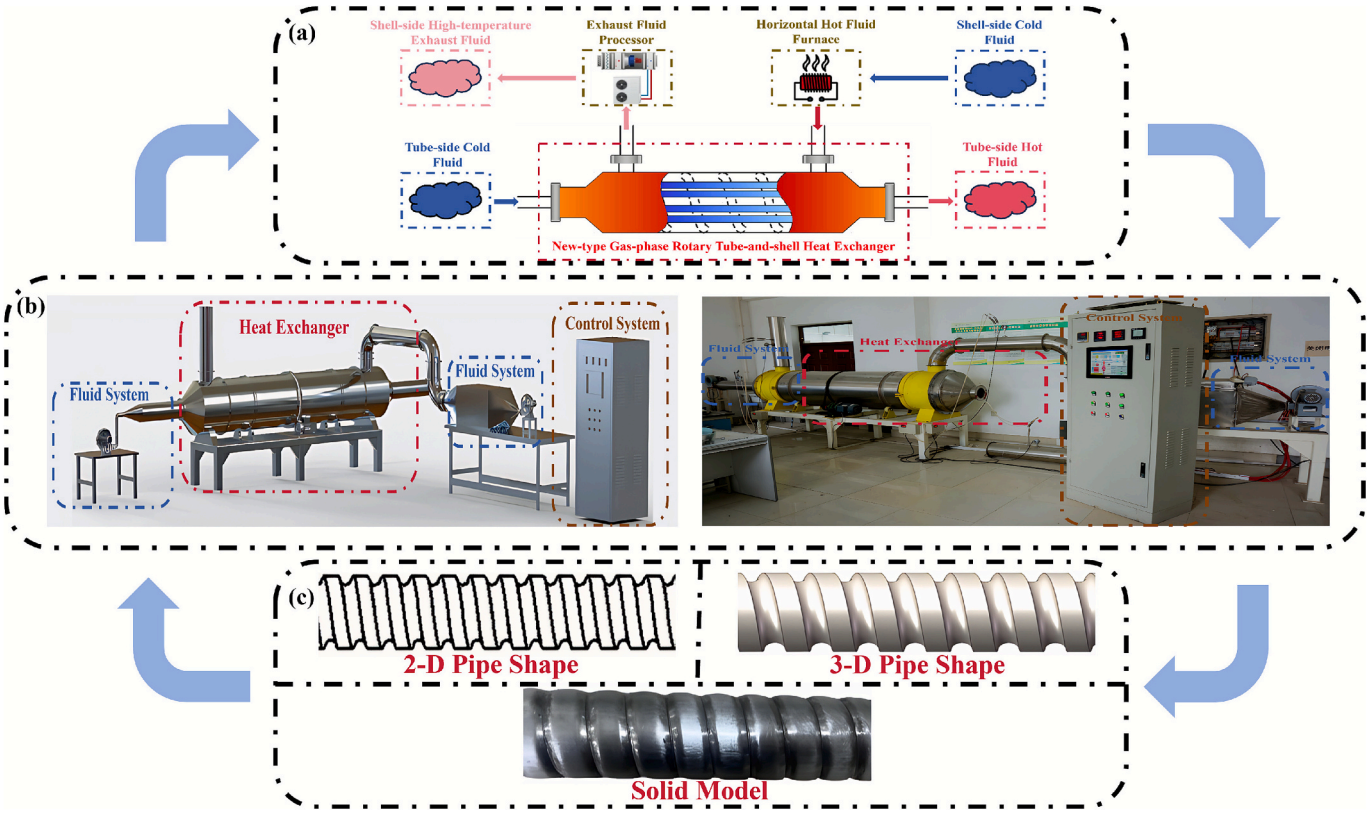


Fig. 1. Schematic diagram of the heat exchanger test platform: (a) operating principle and system structure; (b) 3D and solid models; (c) geometry model of the helical groove tube.

groove tube are listed in Table 1.

The working fluid selected for this study is air at a temperature of 300 K, with its physical properties listed in Table 2.

2.1.2. Governing equations

To investigate the flow field and heat transfer characteristics within the heat exchange tube, it is essential to develop a three-dimensional fluid dynamics model. Under fully developed flow conditions, the mathematical model for convective heat transfer assumes that the temperature distribution of the fluid particles at a given infinitesimal cross-section is independent of the fluid flow. The calculation formula is given as Eq. (1) (Wang et al., 2025). In the equation, $T_{w•m}$ is the average wall temperature at the fluid cross-section, T_b is the average temperature of the fluid particle at the infinitesimal cross-section, T_L is the local fluid temperature, and Z is the coordinate along the flow direction of the particle.

$$\frac{\partial}{\partial_z}\left(\frac{T_{w•m}-T_L}{T_{w•m}-T_b}\right)=0 \tag{1}$$

The numerical solution is based on the Navier-Stokes equations and the Transition SST turbulence model. Fluid flow and heat transfer of any form follow the governing equations, including the continuity Eq. (2), the momentum Eq. (3), and the energy Eq. (4) (Wang et al., 2025). In these equations, the indices i, j, k refer to the coordinate directions.

$$\frac{\partial \rho}{\partial t}+\frac{\partial(\rho u_i)}{\partial x_i}=0 \tag{2}$$

$$\frac{\partial(\rho u_i)}{\partial t}+\frac{\partial(\rho u_i u_j)}{\partial x_j}=-\frac{\partial P}{\partial x_i}+\frac{\partial}{\partial x_j}\left[\mu\left(\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}-\frac{2}{3}\delta_{ij}\frac{\partial u_j}{\partial x_j}\right)\right]+\frac{\partial}{\partial x_j}\left(-\rho\overline{u_i' u_j'}\right) \tag{3}$$

$$\frac{\partial}{\partial x_i}\left[u_i\left(\rho\left(c_p T-\frac{P}{\rho}+\frac{u^2}{2}\right)+P\right)\right]=\frac{\partial}{\partial x_i}\left[\frac{\partial T}{\partial x_j}\left(\lambda+\frac{e_p u_t}{Pr_t}\right)+\mu\mu_i\left(\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}-\frac{2}{3}\delta_{ij}\frac{\partial u_j}{\partial x_j}\right)\right] \tag{4}$$

2.1.3. Boundary conditions and numerical methods

For the fluid flow and heat transfer processes, the following simplifications and assumptions are made in this study: (1) The fluid is assumed to be incompressible and Newtonian, with constant physical properties for both the fluid and the pipe material, and the effects of gravity are neglected; (2) The fluid is assumed to be in a turbulent state, and the flow is fully developed; (3) The fluid moves in a forced convection manner, with the effects of buoyancy and internal heat sources, as well as radiation heat loss, ignored.

The boundary conditions for the model are set as follows: At the inlet, velocity and temperature boundary conditions are applied, with a constant inlet temperature, T_{inlet}=300 K; at the outlet, pressure boundary conditions are applied, with the outlet pressure relative to atmospheric pressure, P_{outlet}=0 Pa; at the pipe walls, conditions of constant wall temperature, no-slip, and impermeability are assumed, with a constant wall temperature, T_{wall}=333 K.

The k-ω SST turbulence model is employed to separately address the flow characteristics near the wall and far from the wall. The coupled pressure–velocity solver is used for discretization, enabling effective resolution of high-gradient boundary layer structures caused by intense turbulence effects. The governing equations are discretized using a steady-state three-dimensional format with the finite volume method. The continuity, momentum, and energy equations are discretized using a second-order upwind scheme, and the relaxation factors are set to the default values used in Fluent software. The convergence criterion for the energy equation residual is 10^{-8}, while the residual convergence criterion for other equations is 10^{-6}, ensuring computational accuracy.

Table 1
Structural dimensions.

Number of Grooves n	Inner Diameter of the Groove Tube d_{in}/mm	Outer Diameter of the Groove Tube d_{out}/mm	Groove Width w/mm	Groove Depth e/mm	Pitch p/mm	Helix β/°
1	36	40	10	5.0	20	8.62

Table 2
Fluid properties parameters.

Working Medium	Fluid Density ρ/kg·m^{-3}	Specific Heat Capacity c_p/J (kg·K)$^{-1}$	Thermal Conductivity λ/W·(m·K)$^{-1}$	Viscosity Coefficient μ/N•s•m^{-2}
Air	1.260	1.005	2.475e-2	1.745e-5

2.1.4. Dimensionless parameters

Commonly used heat transfer performance evaluation indices for heat exchanger tubes include the Nusselt number (Nu) and Darcy friction factor (f). The Nu value is used to assess the heat transfer capability of the fluid, while the f value reflects the flow resistance. The Reynolds number (Re), as a dimensionless flow characteristic parameter, is also an important feature parameter. However, due to the small value of f, this study uses the pressure drop (ΔP) as a substitute metric to more accurately assess the fluid's flow characteristics. The expressions for the Re, Nu, and ΔP values are given by Eqs. (5), (6), and (7) (Wan et al., 2024; Diao et al., 2019; Tian et al., 2024; Zheng et al., 2025). The thermal efficiency to pressure drop ratio, R, serves as a comprehensive performance indicator, directly reflecting the relationship between the Nu and ΔP values. Its expression is shown in Eq. (8) (Wan et al., 2024; Diao et al., 2019; Tian et al., 2024; Zheng et al., 2025).

$$Re=\frac{\rho v d_e}{\mu} \tag{5}$$

$$Nu=\frac{q d_e}{(T_{wall}-T_{fluid})\lambda} \tag{6}$$

$$\Delta P=P_{outlet}-P_{inlet} \tag{7}$$

$$R=Nu/\Delta P \tag{8}$$

2.1.5. Grid independence analysis

An unstructured tetrahedral mesh was used for the numerical model, and grid independence was evaluated to eliminate the influence of grid resolution on the simulation results, as shown in Table 3. Under the condition ofRe = 17463, the grid density was progressively increased, with the number of elements increasing by approximately 50 % at each step (from 0.55 million to 0.83 million, 1.24 million, and 1.86 million). The fluid temperature T^n_{fluid} and inlet mass flow rate $\dot{m}^n_{inlet}$ were recorded for each mesh density. Subsequently, the relative error between the results of successive grid refinements was calculated. The results indicated that with increasing grid density, the value of T^n_{fluid} gradually decreased, and $\dot{m}^n_{inlet}$ also showed noticeable variation. When the number of grid elements increased from 0.55 million to 1.86 million, the decrease in T^n_{fluid} became progressively smaller, with the rate of change at each step being 1.42 %, 0.45 %, and 0.03 %, respectively. Similarly, the variation in $\dot{m}^n_{inlet}$ was significantly reduced, with the change rates being 4.26 %, 0.84 %, and 0.40 %. When the relative error was less than 0.50 %, the results were considered to meet the grid independence criterion. To ensure the accuracy of the numerical simulation, the grid resolution should not be lower than 1.86 million elements. After local

Table 3
Grid independence test.

Number of Mesh Elements	T^{n}_{fluid}/K	$\left\|(T^{1.5n}_{fluid} - T^{n}_{fluid})/T^{n}_{fluid}\right\|$/%	$\dot{m}^{n}_{inlet}$/kg·s^{-1}	$\left\|\left(\dot{m}^{1.5n}_{inlet} - \dot{m}^{n}_{inlet}\right)/\dot{m}^{n}_{inlet}\right\|$/%
55e4	321.4876	–-	0.002349	–-
83e4	316.9413	1.42	0.002249	4.26
124e4	315.5198	0.45	0.002268	0.84
186e4	315.4398	0.03	0.002259	0.40

refinement, the total number of grid elements for the numerical model was approximately 2 million.

This study focuses on analyzing the fluid flow characteristics in the near-wall region. Therefore, a local grid refinement is applied to this area, creating a fine mesh zone. Considering the complex velocity and temperature distributions at the inlet and outlet, as well as the significant variations in the boundary layers of both the flow and thermal fields, this region also requires mesh refinement to form a dense grid zone. The base grid size for the model is set to 0.1 mm, which is suitable for the primary mesh. Boundary layer meshes are used on the wall, with a total of 5 mesh layers. The thickness of the first mesh layer is 0.01 mm, and the growth factor is set to 1.15, resulting in the thickness of the final mesh layer near the wall being 0.0175 mm. The numerical model is illustrated in Fig. 2. The parameter y^+ is a dimensionless quantity that characterizes the distance of the viscous sublayer near the wall (Wang et al., 2025), and its definition is given by Eq. (9), where Δy represents the distance from the first mesh layer to the wall, and τ_w denotes the wall shear stress. Fig. 2(a) illustrates the y^+ distribution on the inner wall, with values ranging from 0.8 to 4.81. The maximum and mean y^+ values are 4.81 and 2.17, respectively, both conforming to the model's near-wall mesh quality criteria. After local refinement, the total number of grids in the model is approximately 2 million (Fig. 2b). The mesh exhibited excellent quality, with a minimum orthogonality of 0.3107, a maximum skewness of 0.47, a peak aspect ratio of 22.68, an adjacent-cell volume ratio of 1.24 and a growth rate of 1.15. All of these parameters satisfy the boundary-layer mesh generation requirements (orthogonality $\geq$ 0.3, skewness $\leq$ 0.5, volume ratio $\leq$ 1.3), thereby ensuring both the stability and accuracy of the numerical simulations.

$$y^+ = \frac{\Delta y \rho}{\mu} \bullet \sqrt{\frac{\tau_w}{\rho}} \tag{9}$$

2.1.6. Numerical model validation

To verify the accuracy of the numerical model, the k-ω SST turbulence model was validated against theoretical data. Within the *Re* range of approximately 6000 to 34000, the simulated *Nu* values were compared with those predicted by the *Gnielinski* correlation (Eq. (10) (Zhu and Zhang, 2014; Yang et al., 2022; Hu et al., 2023), and the validation results are shown in Fig. 3. The results demonstrated that the average error in the *Nu* values was 2.08 %, with a high degree of agreement between the simulation results and the theoretical predictions, thereby confirming the validity of the model.

$$\begin{cases} Nu = 0.0214\left(Re^{0.8} - 100\right)Pr^{0.4}\left[1 + \left(d_e/L_{\text{eff}}\right)^{2/3}\right]\left(T_m/T_{\text{wall}}\right)^{0.45} \\ 0.6 < Pr < 1.5, 0.5 < T_m/T_{\text{wall}} < 1.5, 2300 < Re < 10^6 \end{cases} \tag{10}$$

2.2. Data analysis methods

2.2.1. Significance analysis methods

Significance analysis is used to reveal the degree of correlation between parameters. In this study, the Spearman rank correlation coefficient and one-way Analysis of Variance (ANOVA) methods were employed to examine the relationships between structural parameters (tube diameter d_{in}, groove depth e, pitch p), operating parameters (air velocity v, inlet fluid temperature T_{inlet}, wall temperature T_{wall}), and the *Nu* as well as pressure drop ΔP values. A comprehensive experimental design based on six parameters was implemented, generating 729 sets of data (27 combinations of geometric parameters × 27 combinations of operating parameters) using CFD software. These data were then subjected to significance analysis.

2.2.2. Quantitative analysis methods

Based on the empirical formulas for structural parameters from the literature (Wan et al., 2024), a new empirical formula (Eq. (11) was proposed by combining different structural and operational parameter combinations. A genetic algorithm was utilized for data fitting (Wang et al., 2025), with the mean square error (MSE) between the predicted value ($Nu_{predict}$) and the true value (Nu_{ture}) serving as the fitness function. The result was the development of a new empirical formula to describe the relationship between the parameters d_{in}, e, p, v, T_{inlet}, and T_{wall} and the *Nu* value. Since the *Nu* is a dimensionless parameter, the velocity v was replaced by the *Re*.

$$Nu = a \bullet Re^m \bullet Pr^n \bullet (e/d_{in})^{a_1} \bullet (p/e)^{m_1} \bullet (T_{inlet}/T_{wall})^{n_1} \tag{11}$$

2.2.3. Parameter estimation method

The parameter estimation process includes the following steps: First, the Box-Behnken design is employed to perform experimental design for the six parameters and two response variables. The parameter levels and the design scheme are detailed in Tables 4 and 5, respectively. Next, based on the data obtained from the Box-Behnken design experiments, a response surface regression model is constructed using Workbench software (Kriging model is used as the regression model in this study). Response surface analysis is then conducted to assess the effects of each parameter on the *Nu* and ΔP values (due to the large number of parameters, only the most significant parameters are selected for analysis of their interactions). Finally, the NSGA-II algorithm is applied to optimize the parameters and determine the optimal parameter combination. To ensure the reliability of the analysis results, three sets of repeated simulations are designed (only one set of simulation results is listed in the table below, with an error rate of approximately 0.037 % between the sets).

Based on the Kriging model, this study employs the NSGA-II algorithm with the dual objectives of maximizing the *Nu* and minimizing the ΔP. The optimal parameter combination is selected from the *Pareto* front solution set using a combination of extreme entropy weights and the technique for order of preference by similarity to ideal solution (EWM-TOPSIS method). The NSGA-II algorithm is set with a population size of 200, an iteration count of 500, an initial crossover rate (p_c) of 0.9, and a mutation rate (p_m) of 0.3. The initial λ and β values are set to 0.05 and 1, respectively, and the elite retention strategy size is 10. The fitness function and constraints are described in Eq. (12). Following the formulas in references (Yang et al., 2024) and (Ren and Cao, 2022), the entropy values, weights, and the relative closeness (C_i) of each *Pareto* solution for *Nu* and ΔP are computed, enabling the determination of the optimal parameter combination and its corresponding results.

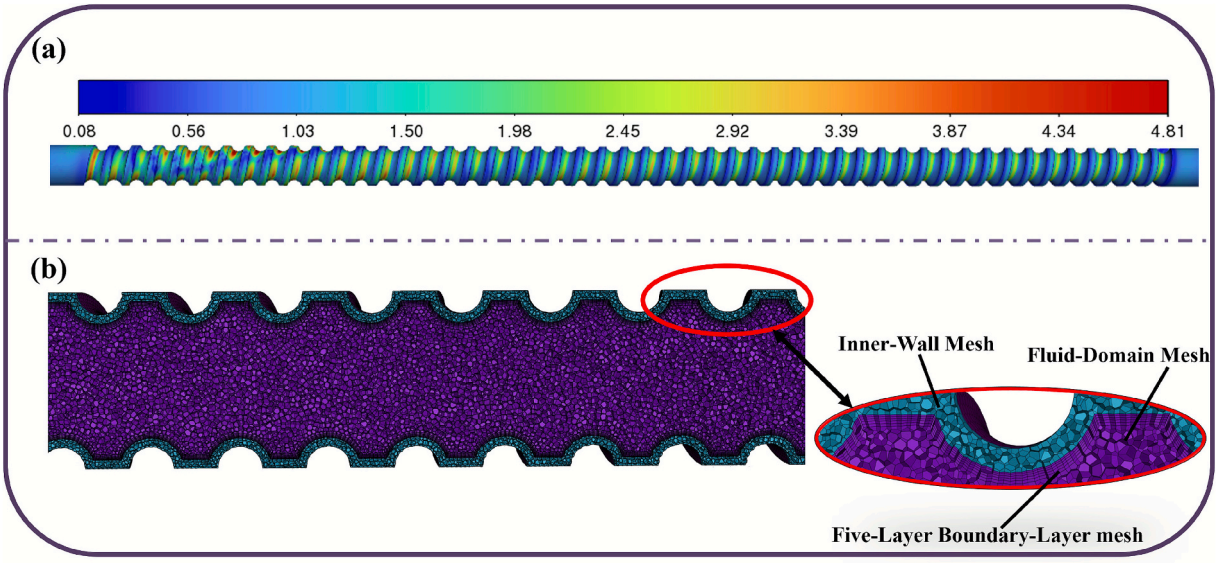


Fig. 2. Numerical model: (a) distribution of y^+ values; (b) mesh division diagram.

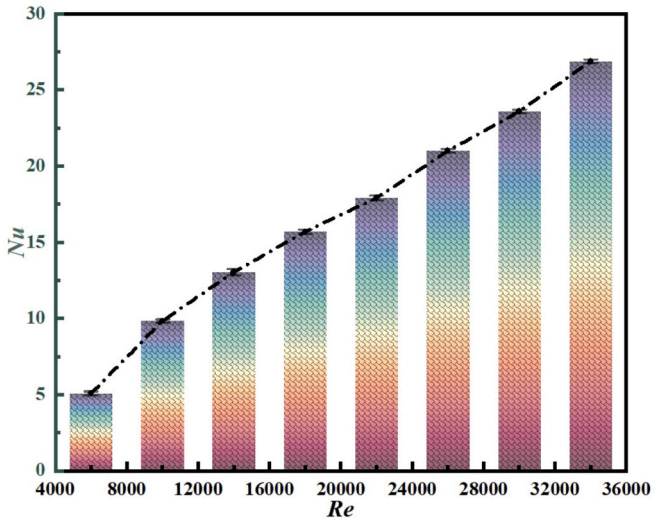


Fig. 3. Verification of the numerical model.

Table 4
Parameter levels table.

Level	Structural Parameters			Operating Parameters		
	d_{in}/mm	e/mm	p/mm	v/m·s^{-1}	T_{inelt}/K	T_{wall}/K
−1	36	2.0	20	2.0	280	333
0	40	3.5	25	6.0	290	343
1	44	5.0	30	10.0	300	353

Note: Based on the content and actual operating conditions presented in references (Wan et al., 2024)- (Zheng et al., 2025), reasonable ranges for each parameter have been selected in this study. Additionally, to facilitate the investigation of heat transfer characteristics, the depth of the groove, denoted as e, is extended to a range of 2.0 to 5.0.

$$\begin{cases} max''Nu'' \\ min''\Delta P'' \\ s.t\begin{cases} 36 \le d_{in} \le 44 \\ 2.0 \le e \le 5.0 \\ 20 \le p \le 30 \\ 2.0 \le v \le 10.0 \\ 280 \le T_{inlet} \le 300 \\ 333 \le T_{wall} \le 353 \end{cases} \end{cases} \tag{12}$$

2.3. Continuous adjoint optimization method

In the current numerical model, the flow field within the pipe is primarily described by the governing equations (*Navier-Stokes* equations), where the state variable u represents the flow field characteristics, the design variable α corresponds to the shape of the inner pipe wall, and the objective function J consists of the Nu and ΔP, as expressed in Eq. (13). The key to this optimization is the calculation of the gradient of the objective function with respect to the design variable, *i.e.*, the sensitivity $dJ/d\alpha$.

$$R(u,\alpha) = 0, J = J(u,\alpha) \tag{13}$$

To enhance the efficiency of sensitivity calculations, the *Lagrangian* function is introduced, where λ represents the *Lagrange* multiplier (*i.e.*, the adjoint variable). Additionally, variational methods are employed to derive the *Lagrangian* function with respect to u, α, and λ as shown in Eq. (14), and setting the derivatives to zero leads to the adjoint Eq. (15). This equation characterizes the evolution of the adjoint field, and by solving the adjoint equation, the *Lagrange* multiplier λ can be obtained.

$$L(u,\alpha,\lambda) = J(u,\alpha) + \lambda^T R(u,\alpha) \tag{14}$$

$$\left(\frac{\partial R}{\partial u}\right)^T \bullet \lambda = -\frac{\partial J}{\partial u} \tag{15}$$

Table 5
Experimental design scheme.

No.	Structural Parameters			Operating Parameters			Evaluation Metric	
	d_{in}/mm	e/mm	p/mm	v/m·s^{-1}	T_{inlet}/K	T_{wall}/K	Nu	ΔP/Pa
1	36	2.0	25	2.0	290	343	4.12	9.31
2	44	2.0	25	2.0	290	343	3.35	6.74
3	36	5.0	25	2.0	290	343	6.36	28.80
4	44	5.0	25	2.0	290	343	5.58	21.35
5	36	2.0	25	10.0	290	343	17.21	245.45
6	44	2.0	25	10.0	290	343	16.96	223.96
7	36	5.0	25	10.0	290	343	30.13	617.89
8	44	5.0	25	10.0	290	343	19.03	512.06
9	40	2.0	20	6.0	280	343	11.90	78.11
10	40	5.0	20	6.0	280	343	20.49	247.71
11	40	2.0	30	6.0	280	343	9.04	58.96
12	40	5.0	30	6.0	280	343	18.33	201.70
13	40	2.0	20	6.0	300	343	7.90	75.11
14	40	5.0	20	6.0	300	343	16.49	245.71
15	40	2.0	30	6.0	300	343	6.14	56.12
16	40	5.0	30	6.0	300	343	15.06	197.03
17	40	3.5	20	2.0	290	333	4.10	12.87
18	40	3.5	30	2.0	290	333	5.12	15.85
19	40	3.5	20	10.0	290	333	20.86	390.18
20	40	3.5	30	10.0	290	333	21.91	388.17
21	40	3.5	20	2.0	290	353	3.17	13.16
22	40	3.5	30	2.0	290	353	8.08	16.01
23	40	3.5	20	10.0	290	353	23.69	391.39
24	40	3.5	30	10.0	290	353	25.04	388.95
25	36	3.5	25	2.0	280	343	6.45	18.28
26	44	3.5	25	2.0	280	343	7.31	12.61
27	36	3.5	25	10.0	280	343	28.77	468.59
28	44	3.5	25	10.0	280	343	26.70	370.44
29	36	3.5	25	2.0	300	343	6.89	18.24
30	44	3.5	25	2.0	300	343	8.91	12.58
31	36	3.5	25	10.0	300	343	16.78	468.57
32	44	3.5	25	10.0	300	343	23.01	370.39
33	40	2.0	25	6.0	280	333	12.87	74.61
34	40	5.0	25	6.0	280	333	18.30	120.79
35	40	2.0	25	6.0	300	333	8.02	74.58
36	40	5.0	25	6.0	300	333	5.88	120.87
37	40	2.0	25	6.0	280	353	16.30	74.70
38	40	5.0	25	6.0	280	353	23.86	120.83
39	40	2.0	25	6.0	300	353	13.17	74.63
40	40	5.0	25	6.0	300	353	8.07	120.91
41	36	3.5	20	6.0	290	333	17.69	147.68
42	44	3.5	20	6.0	290	333	7.39	189.36
43	36	3.5	30	6.0	290	333	15.22	163.93
44	44	3.5	30	6.0	290	333	5.26	148.68
45	36	3.5	20	6.0	290	353	20.90	147.73
46	44	3.5	20	6.0	290	353	9.90	189.42
47	36	3.5	30	6.0	290	353	19.33	163.97
48	44	3.5	30	6.0	290	353	9.20	148.72
49	40	3.5	25	6.0	290	343	18.58	158.36
50	40	3.5	25	6.0	290	343	18.58	158.36
51	40	3.5	25	6.0	290	343	18.58	158.36
52	40	3.5	25	6.0	290	343	18.58	158.36
53	40	3.5	25	6.0	290	343	18.58	158.36
54	40	3.5	25	6.0	290	343	18.58	158.36

The sensitivity calculation formula for this numerical model is given by the Eq. (16). This formula indicates that, by solving the flow and adjoint equations, the sensitivity of the objective function J with respect to the design variable α can be determined.

$$\frac{dJ}{d\alpha} = -\frac{\partial J}{\partial \alpha} + \lambda^T \bullet \frac{\partial R}{\partial \alpha} \tag{16}$$

In Fluent 2024 R2 software, the adjoint optimization process for this numerical model can be divided into the following steps: First, a flow solution is obtained through forward fluid dynamics calculations to solve the flow field. Next, based on the flow field solution, the adjoint equation is solved. Subsequently, using the sensitivity information obtained from the adjoint solver, a sensitivity analysis is performed to calculate the sensitivity of the objective function J with respect to the design variable α. Based on the gradient information, the design variables are updated using the gradient descent method, and the objective function J is iteratively optimized until convergence to the optimal value is achieved.

2.4. PINN method

Due to the difficulty in manufacturing novel heat exchange tubes using traditional rolling processes, 3D printing technology is commonly employed for shaping. However, 3D printing incurs high costs, and the arrangement of metal particles in printed tubes differs significantly from that achieved by rolling, leading to variations in their physical properties. Additionally, heat transfer experiments within the tube flow field cannot be directly conducted on the heat exchanger platform (as shown in Fig. 1). To address this, a physics-informed neural network (PINN) is employed to solve the Navier-Stokes equations and capture the flow and

temperature-field characteristics inside the special-shaped tube, thereby cross-validating the simulation results and enhancing both the reliability of the numerical predictions and the robustness of the optimization conclusions.

The construction process of the PINN surrogate model can be divided into three main steps. Step 1: The helical groove tube cross section in the XY plane is described by the two-dimensional turbulent heat-transfer field $T^{(0)}(x,y,\theta)$, which satisfies the partial-differential operator $g(T)$; Step 2: A neural network is employed to approximate $T^{(0)}(x,y,\theta)$ by mapping the inputs $\{x,y,\theta\}$ to its neurons, producing an initial prediction $t_0(x,y,\theta)$. Automatic differentiation is then applied to compute the gradients of $t_0(x,y,\theta)$ and thereby construct the residual PDE operator $g(t_0)$; Step 3: The PDE residual $\{g(T)-g(t_0)\}$ is adopted as the loss function, and the network parameters – weights w and biases b – are optimized via gradient descent. After iterative training, the network output $t(x,y,\theta)$ constitutes the PINN surrogate model for $T^{(0)}(x,y,\theta)$.

The neurons are fully connected for information transfer, with the output $\{a_1^{l-1},\cdots,a_n^{l-1}\}$ of each layer serving as the input to the next layer's neurons a_j^l. The inputs are multiplied by weights and added to biases, and then passed through an activation function to compute the output. The specific formulation is shown in Eq. (17), where the superscript l represents the layer number, subscripts i and j represent neuron indices, and $\sigma(z)$ is the sigmoid activation function. Through multiple training iterations, the PINN temperature field $t(x,y,\theta)$ is obtained.

$$a_j^l(x,y,\theta)=\sigma\left(\sum_i w_{ji}^l \bullet a_i^{l-1}(x,y,\theta)+b_j^l\right), t(x,y,\theta)=a^L(x,y,\theta;w,b) \tag{17}$$

Considering that the fluid is in a turbulent state, the momentum transfer of the fluid is governed by the NS equations, which are coupled with the k-ω SST turbulence model. In this model, the flow velocity u is governed by the NS equations. Additionally, the turbulence transport equations for velocity u, turbulence kinetic energy k, and turbulence dissipation rate ω need to be solved, as described in Eq. (18). In the equations, $\nabla\bullet\left(\mu_T\left(\nabla u+(\nabla u)^T\right)\right)$ represents the viscous stress divergence term, P_k denotes the turbulent kinetic energy generation term, β^* is a constant in the k-ω model, while α and β are constants related to the dissipation terms in the k-ω model.

$$\begin{cases}\rho\left(\dfrac{\partial u}{\partial t}+u\bullet\nabla u\right)=-\nabla p+\nabla\bullet\left(\mu_T\left(\nabla u+(\nabla u)^T\right)\right)\\ \dfrac{\partial k}{\partial t}+u\bullet\nabla k=\nabla\bullet(\mu_T\nabla k)+P_k-\beta^*\rho\omega k\\ \dfrac{\partial \omega}{\partial t}+u\bullet\nabla\omega=\nabla\bullet(\mu_T\nabla\omega)+\alpha\dfrac{\omega k}{\mu_T}-\beta\omega^2\end{cases} \tag{18}$$

For the heat transfer description, both the convection term and thermal conductivity are considered. The heat equation is given by Eq. (19).

$$\rho C_p u\frac{\partial T}{\partial x}-\lambda\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}\right)=0 \tag{19}$$

Assuming adiabatic conditions with respect to the external environment, the Neumann boundary conditions are applied to the upper and lower wall surfaces, as shown in Eq. (20). In the equation, $(\partial T/\partial y)$ and R represent the vertical temperature gradient and the perpendicular distance from the wall to the central axis, respectively.

$$\frac{\partial T}{\partial y}=0, y=\pm R \tag{20}$$

For the inlet boundary, the known Dirichlet boundary condition T_{in} is applied. The following boundary function $G(x,y,\theta)$ and distance function $D(x,y,\theta)$ are constructed to satisfy the condition, and an extended function $h(x,y,\theta)$ is derived from the boundary and distance functions, as shown in Eqs. (21) and (22), respectively. Here, L is the length, $x\in[0, L]$, and a_L refers to the network output used to construct the solution in the PINN model.

$$G(x,y,\theta)=T_{in}, D(x,y,\theta)=x/L \tag{21}$$

$$h(x,y,\theta)=T_{in}+(x/L)\bullet a_L(x,y,\theta;w,b) \tag{22}$$

The "hard boundary" approach is employed to set the boundary conditions, unifying the boundary loss with the domain loss without relying on empirical penalty coefficients, thus effectively preventing the occurrence of distorted solutions. In addition, the boundary sample data from the turbulent velocity and temperature fields are used for training, and the physical constraints of the k-ω SST turbulence model are incorporated into the PINN surrogate model to ensure that the physical laws of turbulence are correctly embedded into the neural network. The total loss function $Loss_{hard}$ is given in Eq. (23). In this equation, N_t is the total number of domain sampling points, N_B is the total number of boundary sampling points, $\left(x_t^j,y_t^j\right)$ and $\left(x_B^i,x_B^i\right)$ represent the coordinates of the domain and boundary sampling points, respectively, $h^{(1)}$ is the output of the first layer of the network, and C_0 is the boundary condition bias constant.

$$Loss_{hard}=\frac{1}{N_t}\sum_{j=1}^{N_t}\left|f\left[h^{(n)}\left(x_t^j,y_t^j,\theta\right)\right]\right|^2+\frac{1}{N_B}\sum_{j=1}^{N_B}\left|h^{(1)}\left(x_B^i,x_B^i,\theta-C_0\right)\right|^2 \tag{23}$$

3. Results and discussion

3.1. Significance and quantitative analysis

Fig. 4 demonstrates the significance relationships between the parameters d_{in}, e, p, v, T_{inlet}, and T_{wall} with the Nu and ΔP values. According to the Spearman rank correlation analysis shown in Fig. 4(a), a strong correlation exists between the parameters e, v and both the Nu and ΔP values, with correlation coefficients of 0.9799 and 0.9957, 0.9897 and 0.9989, respectively. The correlation coefficients for other parameters are all greater than 0.80. Additionally, a strong correlation (0.9896) between Nu and ΔP values suggests that the heat transfer enhancement due to the helical ribs comes at the expense of increased pressure drop losses. The ANOVA results presented in Fig. 4(a) indicate that as the levels of the parameters change, the variability in Nu and ΔP values increases significantly, and parameters d_{in}, e, p, v, T_{inlet}, and T_{wall} all exert a significant influence on Nu and ΔP values ($p<0.05$). The analysis of Figs. 4(b) and (c) shows that the fitted parameters a, m, n, a_1, m_1, and n_1 are 0.0458, 0.7297, 0.1489, 0.6974, 0.1946, and 0.0043, respectively, with a coefficient of determination R^2 of 0.9867, which fully validates the effectiveness of the empirical Nu correlation. The fitted empirical Nu equation is shown in Eq. (24). In conclusion, a strong correlation exists between the parameters and the heat transfer characteristics.

$$Nu=0.0458\bullet Re^{0.7297}\bullet Pr^{0.1489}\bullet(e/d_{in})^{0.6974}\bullet(p/e)^{0.1946}\bullet\left(T_{\text{fluid}}/T_{\text{wall}}\right)^{0.0043} \tag{24}$$

3.2. Parameter estimation analysis

Fig. 5 presents the response analysis of the groove depth e, air velocity v, and their effects on the Nu and ΔP in a helical groove tube. As the groove depth e and air velocity v increase, both Nu and ΔP exhibit nonlinear variations, demonstrating a positive nonlinear trend. The color of the response variables transitions from blue-green to yellow–red, highlighting their sensitivity to the independent variables. The increase in Nu and ΔP indicates a significant interaction effect

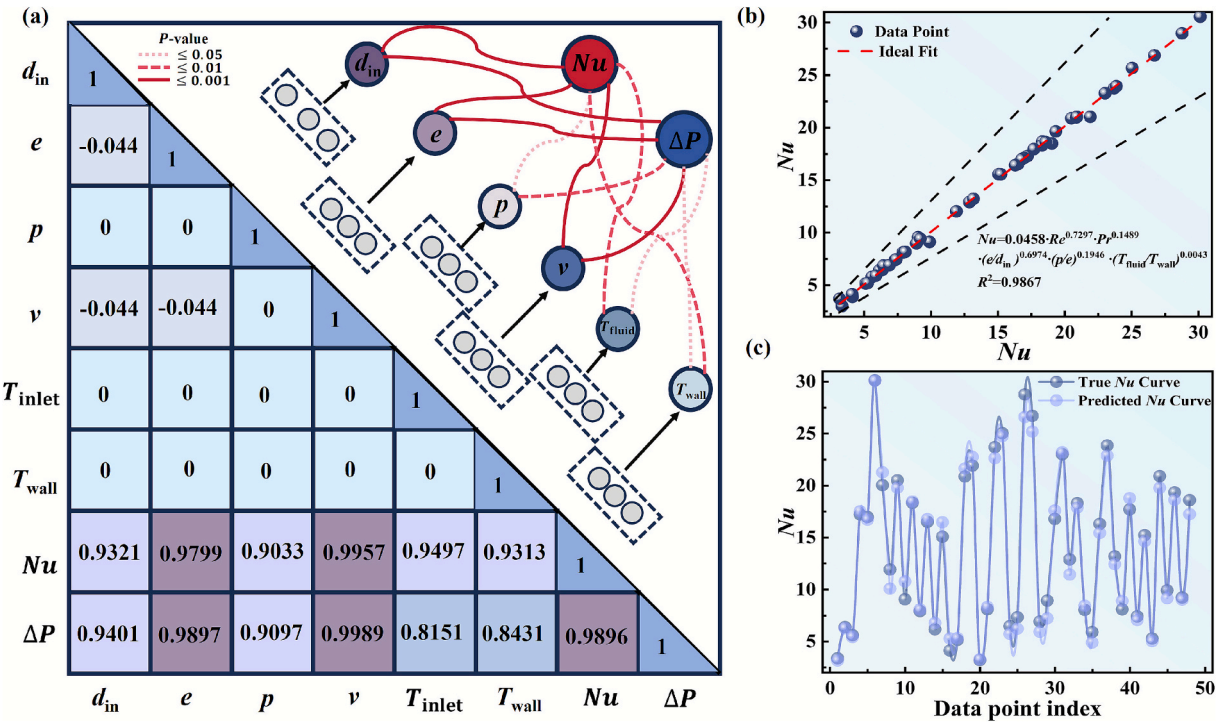


Fig. 4. Significance and quantitative analysis: (a) significance analysis; (b) fitting performance; (c) fitting curve.

between the groove depth e and air velocity v ($p < 0.05$), suggesting that the interaction between these two parameters plays a significant role in regulating heat transfer performance and pressure drop losses (see Figs. 5(a) and (b)). Figs. 5(c) and (d)) respectively show the 2D response analysis and fitting results of Nu and ΔP with respect to groove depth e and air velocity v under different conditions. The results indicate that as the groove depth e increases, the Nu value initially increases but the rate of increase gradually decreases, eventually reaching a decline. This suggests that spiral ribs can effectively enhance heat transfer performance, but a larger groove depth e does not always result in better performance. The ΔP continues to increase, with the rate of increase gradually decreasing, indicating that the influence of groove depth e is most significant within a certain range. As the air velocity v increases, both Nu and ΔP show a continuous increasing trend, with the increase in ΔP following a power-law relationship. The specific mathematical relationships between groove depth e, air velocity v, Nu, and ΔP are obtained through data fitting, as detailed in Eqs. (25) and (26).

$$\begin{cases} Nu = -4.168 + 11.191 \bullet e \pm 1.317 \bullet e^2 \pm 0.00669 \bullet e^3 \\ \Delta P = 234.581 - 280.575 \bullet e - 0.176 \bullet e^{1.600} \end{cases} \tag{25}$$

$$\begin{cases} Nu = \dfrac{4.076 + 2.761 \bullet v}{1 + 0.0143 \bullet v + 0.000699 \bullet v^2} \\ \Delta P = 6.671 \bullet 1.015^{v} \bullet v^{1.719} \end{cases} \tag{26}$$

Fig. 6(a) illustrates the *Pareto* front solution set, which forms a monotonically decreasing curve in the Nu-ΔP plane, revealing the conflicting relationship between Nu and ΔP. This further confirms that increasing the Nu value comes at the cost of increasing the ΔP value, and it is not possible to optimize both objectives simultaneously. Fig. 6(b) presents the results of the EWM-TOPSIS analysis applied to the *Pareto* frontier solution set. Among the algorithm's results, the solution in Group 556 has the highest C_i value ($C_i \approx 0.68$), indicating that this solution is the optimal solution for multi-objective optimization. At this point, the Nu and ΔP values are 19.16 and 67.89, respectively. The corresponding optimal parameters are: d_{in}=38.00 mm, e=3.10 mm, p=25.23 mm, v=5.30 m·s^{-1}, T_{inlet}=280.36 K and T_{wall} = 352.82 K. In Workbench software, updating the optimal parameter combination yields a result with an error of approximately 0.30 %.

3.3. Adjoint optimization analysis

Using the helical groove tube with the optimal parameter combination as the baseline geometry, continuous adjoint optimization was performed, yielding a novel high-efficiency heat-transfer tube. Fig. 7 illustrates the continuous adjoint optimization process and the pre- and post-optimization surface morphologies of the fluid column. Direct analysis of the tube wall is challenging; therefore, the fluid column and its surface have been examined as a surrogate for wall morphology (Fig. 7a). A sensitivity study was performed on the entire inner wall of the helical groove tube as the optimization target (Fig. 7b). Optimization commenced from the baseline Nusselt number (Nu = 19.16) and pressure drop (ΔP = 67.89 Pa), with objectives set to increase Nu by 40 % and decrease ΔP by 40 %. An adaptive solver was employed, with design, adjoint and flow iteration limits of 40, 200 and 500, respectively, and a convergence criterion of 0.005. Boundary deformation was optimized using the CADJ method. After optimization, the maximum and average boundary deformations were 0.27 mm and 0.15 mm, respectively (regions *i-iii* in Fig. 7c are magnified five-fold). The optimized tube achieved Nu = 23.38 and ΔP = 52.36 Pa–corresponding to increases of 22.03 % and decreases of 22.88 %, respectively–falling short of the prescribed targets (Fig. 7d). Geometric constraints limited full attainment of the goals, yet significant improvements in heat-transfer performance and pressure-drop control were realized. Under constant ΔP, a larger performance index R indicates higher energy efficiency (Wang et al., 2025; Stehlik and Wadekar, 2002). Substituting Nu and ΔP into Eq. (8) yielded R = 0.282 Pa^{-1} for the baseline tube and R = 0.447 Pa^{-1} for the optimized tube–a 58.51 % enhancement. Overall, the novel tube demonstrates superior heat-transfer and energy-efficiency characteristics.

Considering the influence of wall morphology, the flow patterns and the distributions of Nu and vorticity magnitude ($|\omega|$) within both tubes are compared in Fig. 8. The magnitude of vorticity reflects vortex strength (Fig. 8b). Compared with the helical groove tube, the optimized

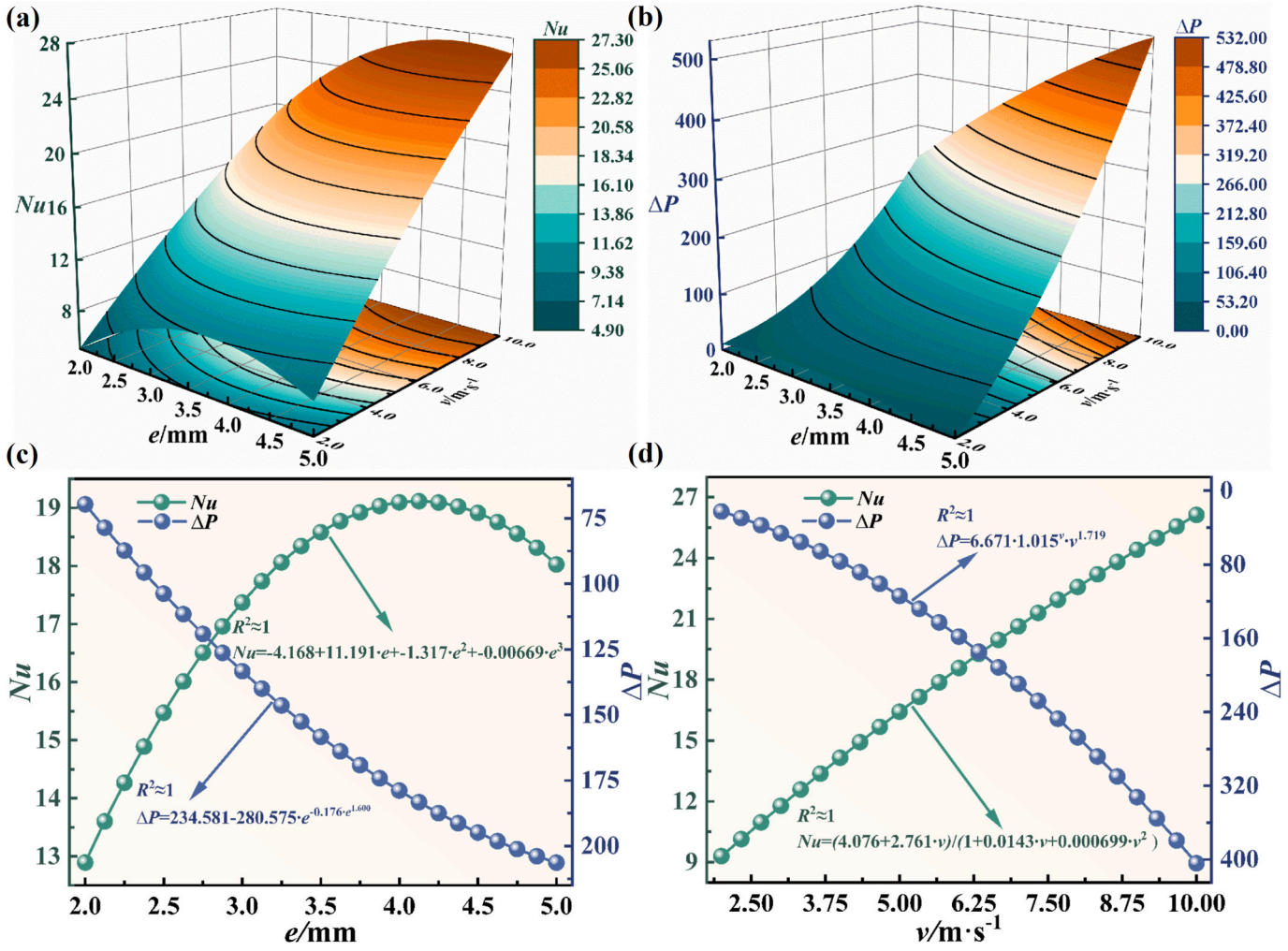


Fig. 5. Response analysis: (a) 3D interactive response of groove depth e and air velocity v on Nu value; (b) 3D interactive response of groove depth e and air velocity v on ΔP value; (c) 2D response analysis of groove depth e on Nu and ΔP values at$v$ = 6 m·s^{-1}; (d) 2D response analysis of flow velocity v on Nu and ΔP values at$e$ = 3.5 mm.

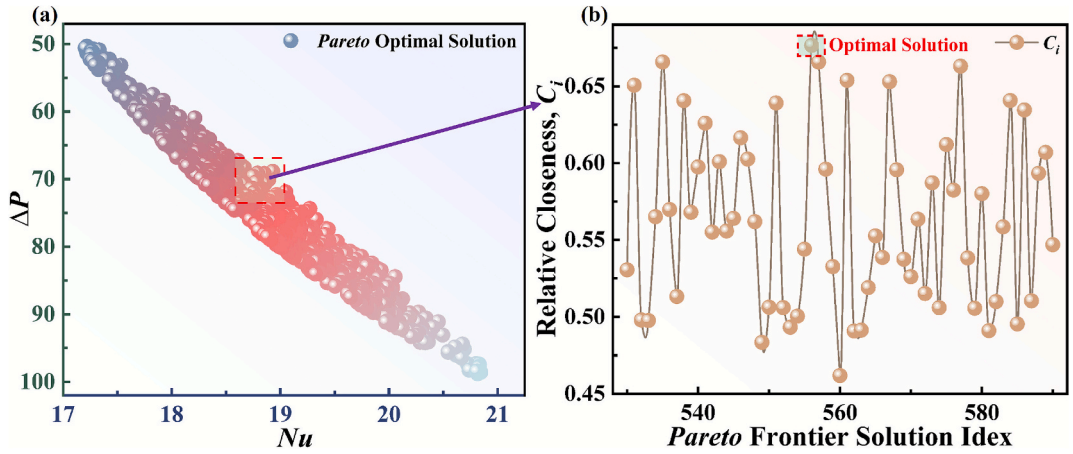


Fig. 6. *Pareto* front solution set analysis: (a) *Pareto* front solution set; (b) relative closeness of *Pareto* solutions C_i.

tube exhibits more chaotic streamlines in regions *A*, *B* and *C*, resulting in larger and more intense vortical structures (Fig. 8a). Fluid trajectories in the optimized tube follow a helical path (region *D*) but remain smoother, indicating lower pressure-drop losses (Fig. 8c). At all measurement locations, both *Nu* and $|\omega|$ exceed those in the baseline tube, with greater fluctuations across points and no clear spatial trend (Fig. 8b). These fluctuations arise partly from local wall morphology changes–amplifying oscillations of *Nu* and $|\omega|$–and partly because the overall inner-wall shape remains largely consistent, preserving the general evolution characteristics of the flow.

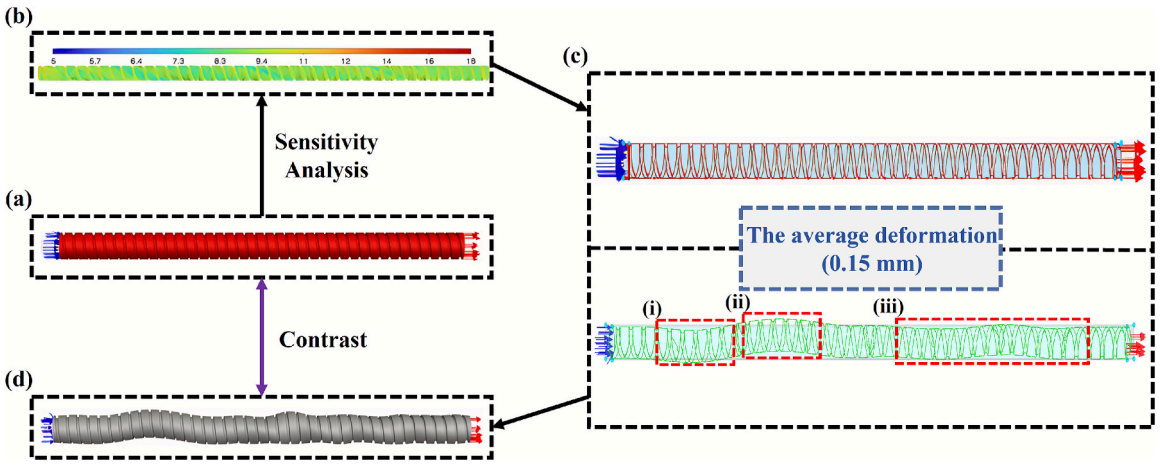


Fig. 7. Continuous adjoint optimization process and morphological evolution of the fluid column before and after optimization: (a) baseline helical groove tube model; (b) sensitivity analysis of *Nu* and ΔP; (c) deformation of the tube-wall boundary; (d) optimized novel heat-exchanger tube model.

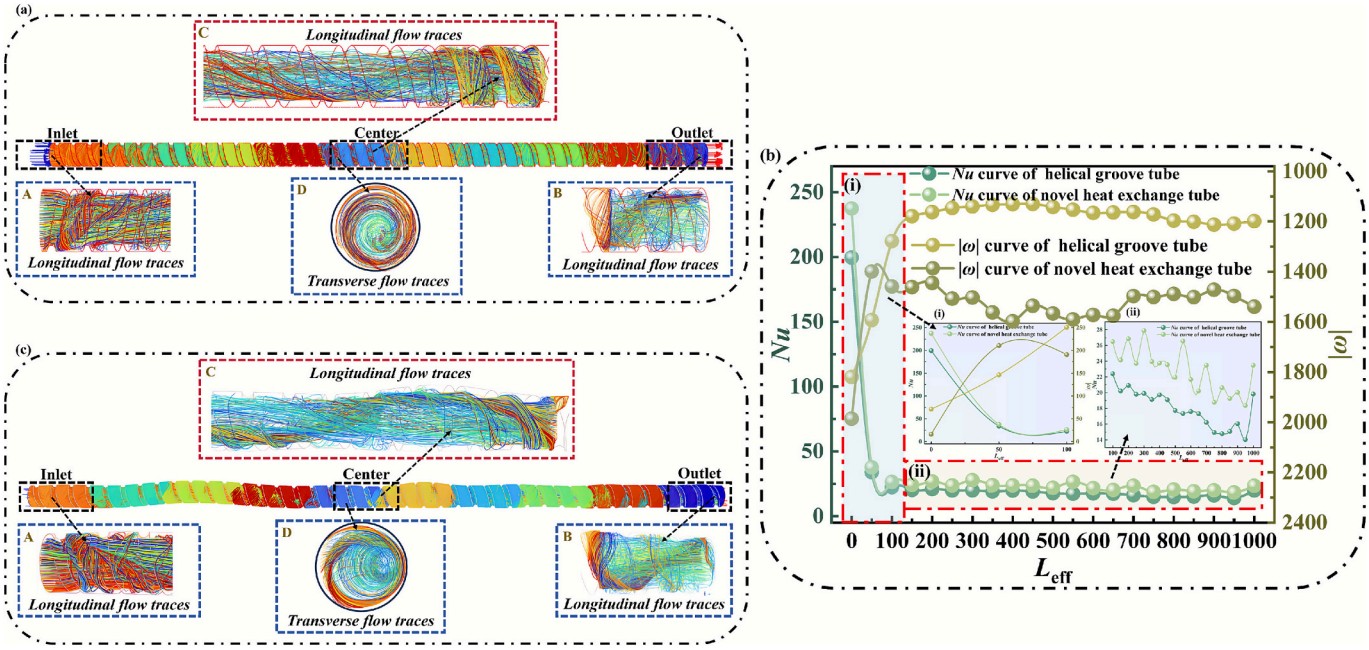


Fig. 8. Flow trajectories and distributions of *Nu* and $|\omega|$: (a) streamlines within the helical groove tube; (b) distributions of *Nu* and $|\omega|$ along the principal flow direction for both tube configurations; (c) streamlines within the novel heat-exchanger tube.

Fig. 9 presents the contours of fluid $|\omega|$ and T_{fluid} values on the XY longitudinal cross-section for two heat exchanger tubes. The results indicate that the novel heat exchanger tube demonstrates a significantly broader range of $|\omega|$ values (see regions *i* and *ii* in Fig. 9a), with an average $|\omega|$ value of approximately 1499.874 s^{-1}, which is roughly 1.25 times that observed for the helical groove tube. This enhancement suggests that the newly designed inner-wall structure effectively strengthens fluid vorticity, thereby substantially improving convective heat transfer characteristics within both the near-wall and mainstream regions. In terms of T_{fluid}, noticeable variations were observed in both the near-wall and mainstream regions of the novel heat exchanger tube (regions *i* and *ii* in Fig. 9b). Specifically, the near-wall region experienced relatively minor temperature fluctuations, whereas the mainstream region exhibited more pronounced changes, with an average T_{fluid} value of 317.06 K, approximately 1.02 times that of the helical groove tube. These results suggest that the novel inner-wall structure reduces the temperature gradient in the near-wall region, consequently increasing the temperature gradient in the mainstream region. This mechanism promotes heat transfer between the mainstream and near-wall fluids, thereby significantly improving overall heat transfer performance. Moreover, the intensified vortices disrupt the boundary layer, enhancing the mixing between high-temperature fluid near the wall and lower-temperature fluid in the mainstream region. Such improved internal thermal energy transfer significantly augments the convective heat transfer characteristics inside the tube, thereby effectively enhancing heat transfer. In addition, compared with the helical groove tube, the increased effective heat-transfer surface area of the novel heat-exchange tube contributes to enhanced heat-transfer performance. In summary, the novel inner-wall structure endows the heat exchanger tube with superior heat transfer performance and reduced pressure-drop losses.

In summary, the heat transfer efficiency of the novel heat exchanger tube exceeds that of the helical groove tube. The key lies in the CADJ method, which continuously improves and adjusts the inner wall morphology (or the fluid column surface morphology) of the helical groove tube, while maintaining the basic structure of the internal helical

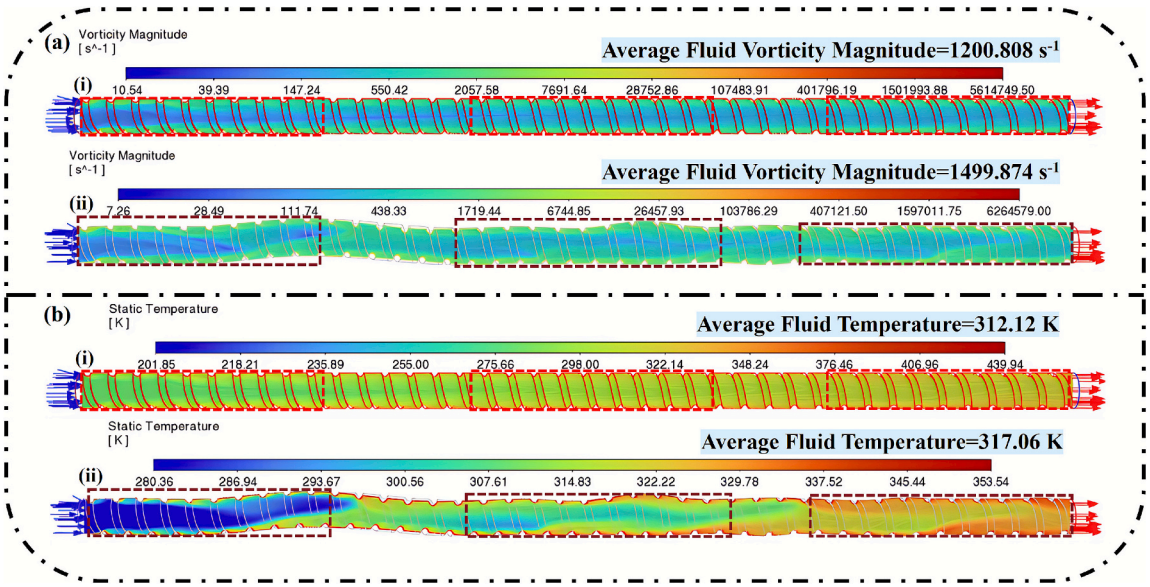


Fig. 9. Distributions of $|\omega|$ and T_{fluid} in the XY cross-section: (a) $|\omega|$ distributions for both tube configurations; (b) T_{fluid} distributions for both tube configurations.

ribs. This leads to the formation of stronger vortex structures while maintaining relatively smooth fluid flow trajectories. This adjustment not only enhances heat transfer performance but also effectively regulates pressure drop losses.

3.4. PINN analysis

In the Python 3.11 and TensorFlow 2.13 environment, a PINN surrogate model was constructed and trained in this study. The model takes three input dimensions $\{x, y, u_{max}\}$, which include the uniformly discretized two-dimensional coordinates (x, y) on a 50 × 100 grid and the parameterized maximum velocity, u_{max} (velocity range: 5.2–5.4 m·s^{-1}), with a total of 50 samples. The model consists of a four-layer fully connected neural network, with 50 neurons per layer and a sigmoid activation function (as shown in Fig. 10). Furthermore, the model's single iteration training time is approximately 15 min, and after 1000 cycles, the loss function reaches a minimum value of 0.071. Under the condition ofu_{max} = 5.30 m·s^{-1}, the temperature field predicted by the model is shown in Fig. 11. As observed in Fig. 11(b), the temperature field of the model closely matches the simulation results, accurately capturing the temperature variation trend. The error distribution in Fig. 11(c) indicates that the error generally remains at the order of magnitude of 10^{-1} within the two-dimensional region, with an average error close to 0.29°C. The error slightly increases in the near-wall regions, with a maximum absolute error of 0.80°C, further demonstrating the high accuracy of the model.

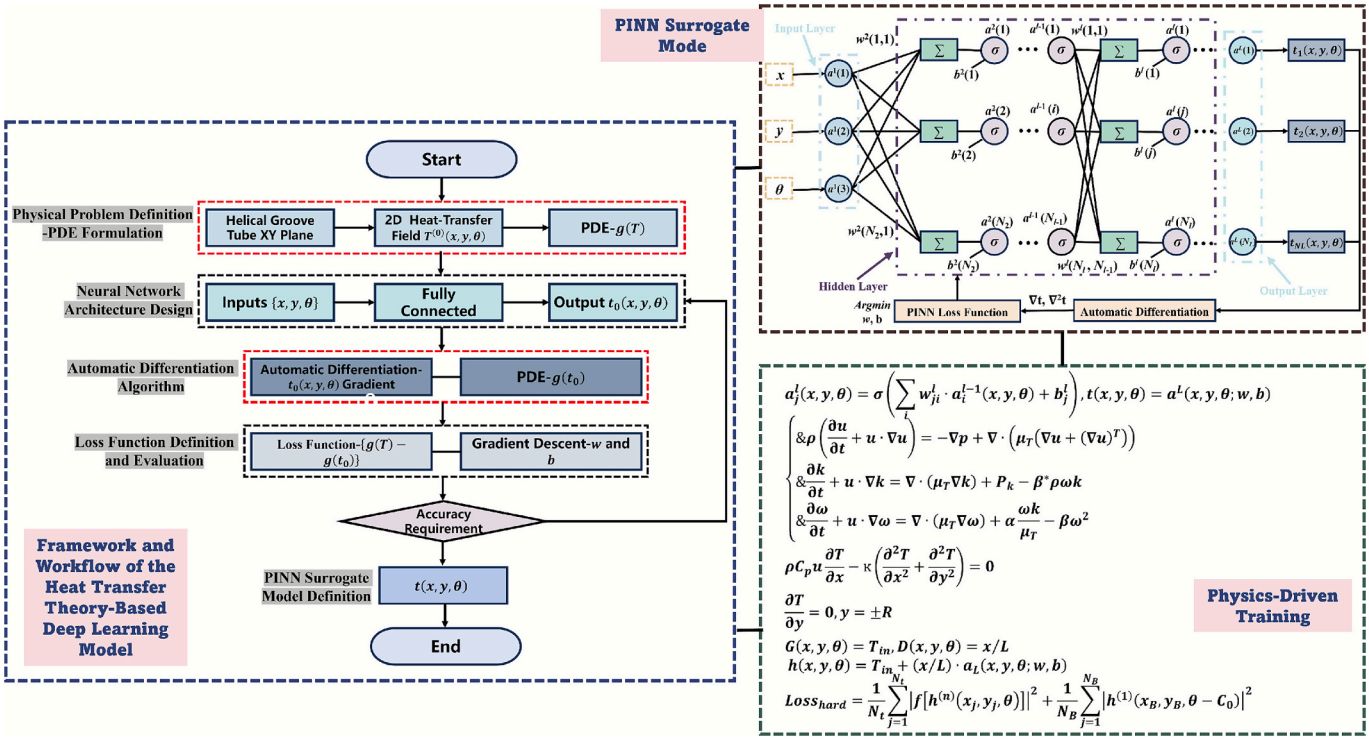


Fig. 10. PINN surrogate model structure diagram.

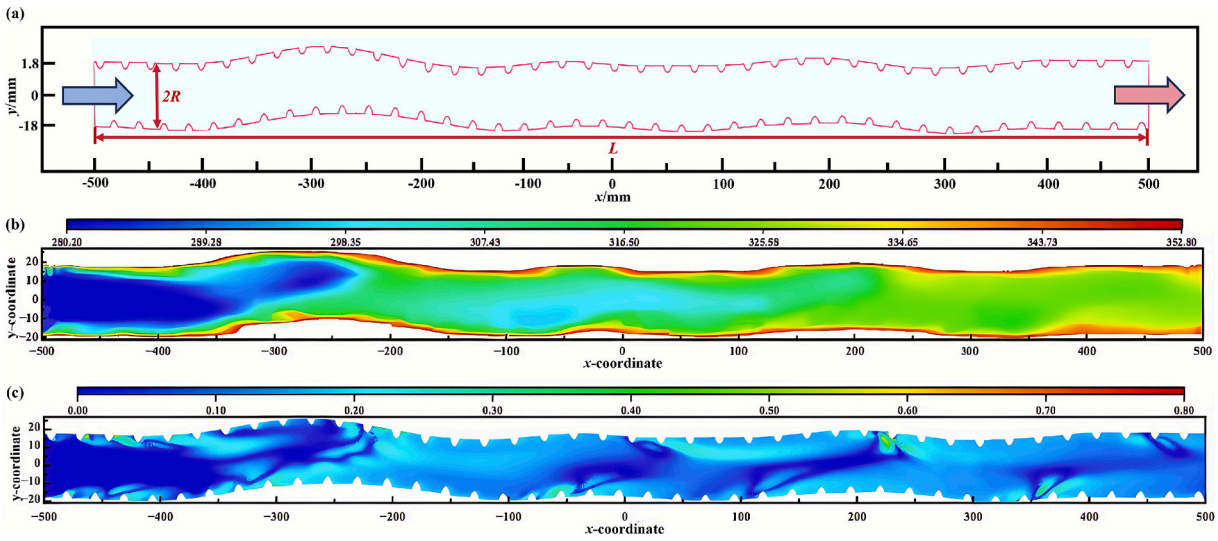


Fig. 11. PINN surrogate model analysis: (a) two-dimensional physical model; (b) temperature field; (c) error distribution.

4. Conclusion

This study focuses on the helical groove tube with the objective of enhancing both heat transfer performance and energy efficiency. By applying PE in conjunction with CADJ, the structural configuration, operating parameters, and wall morphology of the heat transfer tube are systematically optimized, leading to the development of a novel high-efficiency heat exchanger tube. The results demonstrate that:

(1) The groove depth e and air velocity v exhibit strong correlations with both the Nu and ΔP, while the d_{in}, p, T_{inlet}, and T_{wall} also exert significant influence. Following iterative fitting, the empirical correlations yielded parameters a, m, n, a_1, m_1, and n_1 as 0.0458, 0.7297, 0.1489, 0.6974, 0.1946, and 0.0043, respectively, with the R^2 of 0.9867. The corresponding empirical model for Nu is expressed as: $Nu = 0.0458 \bullet Re^{0.7297} \bullet Pr^{0.1489} \bullet (e/d_{in})^{0.6974} \bullet (p/e)^{0.1946} \bullet \left(T_{fluid}/T_{wall}\right)^{0.0043}$.

(2) The optimal operating parameters for the helical groove tube were determined as follows: d_{in}=38.00 mm, e = 3.10 mm, p=25.23 mm, v=5.30 m•s^{-1}, T_{inlet}=280.36 K, andT_{wall} = 352.82 K. Under these conditions, the corresponding Nu, ΔP, and R values were 19.16, 67.89 Pa, and 0.282 Pa^{-1}, respectively. Compared with the helical groove tube, the novel heat exchanger tube achieved a 22.03 % increase in Nu, a 22.88 % reduction in ΔP, and a 58.51 % improvement in R, demonstrating superior overall thermal and energy efficiency.

(3) The enhanced performance of the novel tube can be attributed primarily to the CADJ-based refinement of the inner-wall morphology (or fluid-column surface morphology) of the helical groove tube. While preserving the intrinsic helical rib structure, the optimization intensified vortex generation while maintaining smooth fluid flow trajectories. This approach not only improved heat transfer but also effectively regulated pressure drop losses.

(4) In the absence of experimental validation, the relative error between the temperature field reconstructed by the PINN and the CFD solution remains on the order of 10^{-1}. overall, indicating that using PINN to perform cross-validation and auxiliary assessment of CFD results provides a simple yet efficient alternative approach.

This study presents an innovative and scientifically sound strategy for optimizing heat exchanger tubes, aiming to enhance both heat transfer performance and energy efficiency. This approach holds significant practical value for the optimization and design of heat exchanger tubes. However, there are still some limitations in this research, and future work will focus on further optimizing and refining the related tasks. Additionally, the new heat exchanger tubes will be manufactured using 3D printing technology. Furthermore, the team plans to conduct experimental validation using more complex and extensive particle image velocimetry (PIV) methods, with the goal of applying these findings to the heat transfer field. The relevant results will be submitted to this journal in the future.

CRediT authorship contribution statement

Shuo Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Lin Wan:** Supervision, Resources, Funding acquisition. **Hongchao Wang:** Supervision, Resources, Funding acquisition. **Gang Che:** Supervision, Resources, Funding acquisition. **Yan Li:** Supervision, Resources, Funding acquisition. **Tingbo Du:** Visualization, Software, Investigation. **Chaofan Wang:** Visualization, Software, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was supported by the Natural Science Foundation of Heilongjiang Province under the project number LH2022E098, respectively.

Data availability

All research data are included in the manuscript, with no omissions.

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